

Default Bayes and prediction problems with global-local shrinkage priors

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Overview

- Goal 1: Argument for global-local priors in default Bayes analysis for **low-dimensional functions of high-dimensional normal means**.
 - Efron Problems: sum of squares, maximum, product and ratio of normal means.
- Goal 2: To quantify the **prediction risk** for *global* and *global-local* shrinkage regressions.
 - Stein's unbiased risk estimate (SURE) for global-local shrinkage regression.

Joint work with Jyotishka Datta (Arkansas); Yunfan Li (Purdue); Nick Polson and Brandon Willard (Chicago Booth). Supported by NSF Grant DMS-1613063.

Global-local (g-l) priors

- Consider the sparse “normal means” model $(y_i \mid \theta_i) \sim \mathcal{N}(\theta_i, 1)$ for $i = 1, \dots, n$; such that $\#(\theta_i \neq 0) \leq p_n$ with $p_n = o(n)$.
- Carvalho, Polson and Scott (2010) introduced “global-local” normal scale mixture priors for sparsity

$$(\theta_i \mid \lambda_i, \tau) \sim \mathcal{N}(0, \lambda_i^2 \tau^2); \quad \lambda_i \sim p(\lambda_i); \quad \tau \sim p(\tau).$$

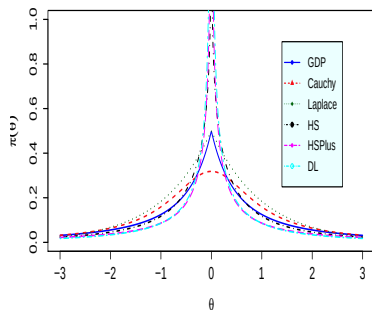
- The “global” term τ should provide substantial shrinkage towards zero.
- The “local” λ_i terms should have heavy tails so that “signals” are not shrunk too much. One option is $p(\lambda_i) \propto (1 + \lambda_i^2)^{-1}$, which induces the “horseshoe prior” on θ .

Some examples of global-local priors

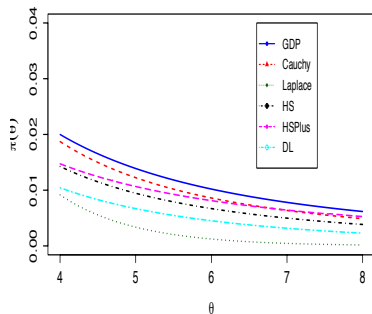
- The horseshoe prior (Carvalho, Polson and Scott, 2010, Biometrika).
- The horseshoe+ prior (Bhadra et al., 2016, Bayesian Anal.).
- The hypergeometric inverted-beta prior (Polson and Scott, 2010, Bayesian Anal.).
- The generalized double Pareto prior (Armagan, Dunson and Lee, 2013, Stat. Sinica).
- The three parameter beta prior (Armagan, Dunson and Clyde, 2011, NIPS).
- The Dirichlet-Laplace prior (Bhattacharya et al., 2015, JASA).

Some examples of global-local priors

- The order of peakedness near zero: $HS+ \approx DL > HS > GDP = Laplace > Cauchy$



- The order of tail heaviness: $GDP > Cauchy > HS+ > HS > DL > Laplace$



Some properties of g-I priors

- [Carvalho et al. \(2010, Biometrika\)](#): showed the the K-L distance between the estimated and the true predictive densities decreases at a super-efficient rate for the horseshoe.
- [Datta and Ghosh \(2013, Bayesian Anal.\)](#): proved that the decision rule induced by the horseshoe estimator is asymptotically Bayes optimal for multiple testing under 0-1 loss.
- [van der Pas, Kleijn and van der Vaart \(2014, EJS\)](#): showed the horseshoe estimator is minimax in ℓ_2 up to a constant.

Beyond the normal means: Efron problems with non-informative priors (Efron, 1973, JRSSB)

Professor BRADLEY EFRON (Stanford University): The question of just what constitutes an uninformative prior in a multiparameter situation becomes ever more vexing, helped along now by the authors' very provocative counterexamples. I only hope that readers will not misread this paper as saying that all is well as long as improper priors are avoided. Suppose we have 100 unknown parameters $\theta_1, \theta_2, \dots, \theta_{100}$ and data $x_1, x_2, \dots, x_{100}$, where $x_i \sim N(\theta_i, 1)$, independently given the $\{\theta_i\}$. We may try to represent our lack of prior information on the θ_i by giving them independent $N(0, A)$ priors where A is enormous, say 10^{1000} . This looks uninformative enough, being virtually equivalent to a uniform prior over E^{100} for most purposes, but like the uniform prior it is actually much too informative in some ways.

For example, suppose we wish to estimate $\xi = \sum_{i=1}^{100} \theta_i^2$, and observe that the 100 x_i values have sum of squares 200. The *a posteriori* mean of ξ given the data are almost exactly 300 in this case, as opposed to the much more reasonable unbiased estimate $\hat{\xi} = 100$, which has estimated standard deviation 25. Our "uninformative" prior has completely overwhelmed the considerable amount of information in the data! This is because it gives ξ a marginal prior density proportional to ξ^{49} (to a close approximation, for $\xi < 10^{999}$), which is heavily weighted against small values of ξ .

We can correct this by giving A itself a diffuse prior, say with density proportional to $(A+1)^{-2}$, instead of a large fixed value, in which case ξ will have marginal prior density approximately proportional to $(\xi+100)^{-2}$, and the *a posteriori* mean of ξ will always be close to the m.l.e. or to the unbiased estimate. Unfortunately this new uninformative prior is quite informative in its own right. For example, if we wish to estimate $\mu = \max \{\theta_i\}$ and observe $x_1, x_2, \dots, x_{99}$ to have nearly a $N(0, 1)$ histogram while $x_{100} = 10$, then the *a posteriori* expectation of μ will be close to 5. It is obvious that 10 is a much more sensible estimate in this case.

Why are statisticians interested in uninformative priors? Because they connect Bayesian and frequentist methods, because they offer an "objective" form of Bayesian theory and because they are so convenient for dealing with complicated situations, particularly those involving nuisance parameters. In the 100 parameter problem for instance, a truly uninformative prior, if it existed, would in principle provide a sensible answer to *every* question one could ask about the parameters, both before and after the data were observed. It is worth looking for such a powerful weapon, but sobering to have pointed out that even in much simpler situations the proposed candidates have undesirable properties.

Efron's problems

- Suppose $\psi = \sum_{i=1}^{100} \theta_i^2$ is the parameter of interest
- We observe $\sum_{i=1}^{100} y_i^2 = 200$.
 - Intuitively $\hat{\psi}$ ought to be 100 with a standard deviation of 25.
 - Posterior mean under $\theta_i \stackrel{\text{ind}}{\sim} \mathcal{N}(0, A)$ with huge A is 300.
Is $\theta_i \stackrel{\text{ind}}{\sim} \mathcal{N}(0, A)$ with $A \rightarrow \infty$ “non-informative”?
- What is non-informative for estimating θ_i s is actually very informative for estimating $\psi = \sum_{i=1}^{100} \theta_i^2$.

$\psi = \sum_{i=1}^{100} \theta_i^2$: normal and horseshoe priors

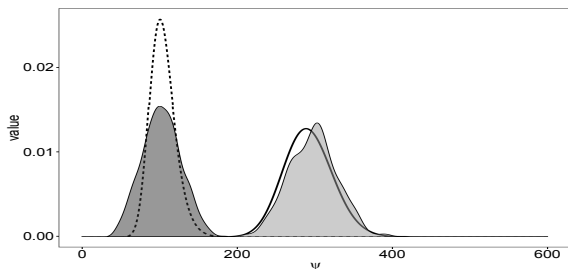


Figure : Posterior under half-Cauchy and $\mathcal{N}(0, 300)$ priors.

True $\psi = 100$. Horseshoe posterior concentrates in the correct region.
Normal prior **wrong!**

Efron's solution and a resultant problem

Efron: Don't fix A at a large value. Instead, diffuse half-Cauchy prior $p(A) \propto (A + 1)^{-1}$.

- The posterior mean of $\psi = \sum \theta_i^2$ is now essentially the James-Stein estimate - good for dense θ .

BUT! suppose parameter of interest is $\phi = \max \theta_i$ and $y_{\max} = 10$.

- Efron points out that posterior estimate of ϕ with half-Cauchy prior on A will be 5 while 10 is much more reasonable.

Cause: JS global shrinkage shrinks everything, small and large!

Our proposal (Bhadra et al., 2016, Biometrika)

- Use Global-local priors (e.g., horseshoe and horseshoe+).
 - $\psi = \sum \theta_i^2$ (sum of squares)
 - $\psi = \max \theta_i$ (max)
 - $\psi = \theta_1 \theta_2$ (product)
 - $\psi = \theta_1 / \theta_2$ (ratio or Fieller-Creasy).
- The **local** heavy-tailed λ_i terms leave large signals un-shrunk, even for nonlinear functions!
- The **global** term helps shrink the noise components, even for nonlinear functions!

A key property: regular variation

Key property: half-Cauchy (Gelman) has regularly-varying tails.

- Regular variation is closed under many nonlinear transformations (including four on the previous slide).
- The regularly varying tails of θ_i s translate to regularly varying tails for the prior of ψ .
- Since the likelihood is light-tailed (normal), the heavy tailed priors on ψ help in non-informative analysis (Dawid, 1973).

Results: candidate priors for the Efron problems

We compare the following priors

- Global-local shrinkage priors, namely, the horseshoe and the horseshoe+ priors.
- Laplace or double-exponential prior:

$$p(\lambda_i^2 \mid \tau^2) = (2\tau^2)^{-1} \exp\{-\lambda_i^2/2\tau^2\},$$
$$\tau^2 \sim \text{IG}(1/2, 1/2).$$

- Vague normal prior, that is, $\theta_i \sim \mathcal{N}(0, \sigma^2 = 300)$.
- Pure-local shrinkage prior, and the pure-global shrinkage priors, by taking $\tau = 1$ or, $\lambda_i = 1$, for all $i = 1, \dots, p$.
- Reference priors (when they exist).

Alternatives: spike-and-slab Lasso, ...

Results: sum of squares problem

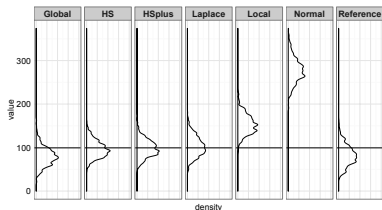


Figure : $A = 10$ and $q_p = 1$.

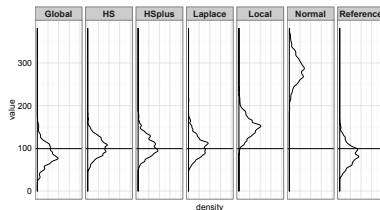


Figure : $A = 5$ and $q_p = 4$.

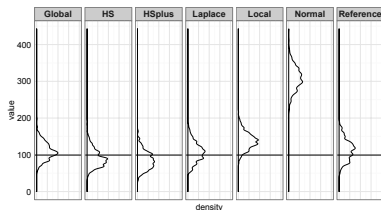


Figure : $A = 1$ and $q_p = 100$.

Figure : Posterior densities of $\psi = \sum_{i=1}^{100} \theta_i^2$, q_p is the number of non-zero means and A is the magnitude. The horizontal line at true $\psi = 100$.

Results: maximum problem

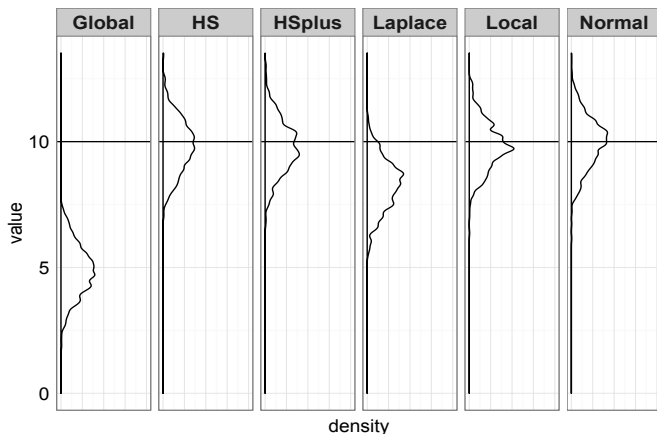


Figure : Posterior densities for $\psi = \max \theta_i$, for $(y_i | \theta_i) \sim \mathcal{N}(0, 1)$, $i = 1, \dots, 99$ and $y_{100} = 10$. The horizontal line is at $y_{\max}$.

Results: product and ratio problems

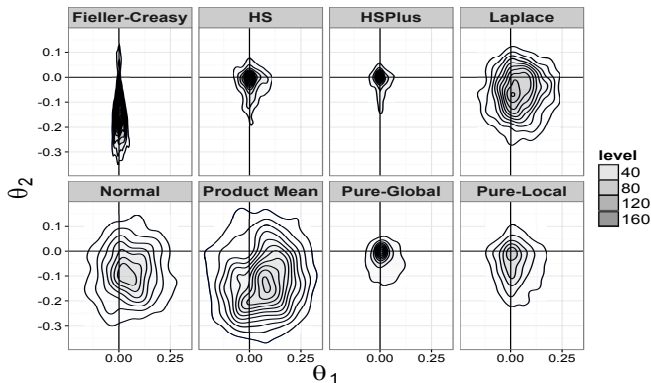


Figure : Two-dimensional contour plots of $p(\theta_1, \theta_2 | y)$ for the product mean and ratio of two means (Fieller-Creasy) problems. True $\theta_1 = \theta_2 = 0$.

G-I priors in orthogonalized high-dimensional regression (Bhadra et al., 2016, arXiv:1605.04796)

- Consider the high-dimensional regression model with $p > n$

$$y = X\beta + \epsilon,$$

where $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$ and $\epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$.

- Let $X = UDW^T$, $\text{Rank}(D) = n$ where $D = \text{diag}(d_i)$ with $d_1 \geq \dots \geq d_n > 0$.
- Define $Z = UD$ and $\alpha = W^T \beta$.
- Then the regression problem can be reformulated as:

$$y = Z\alpha + \epsilon.$$

Shrinkage regression estimates as posterior means (Frank and Friedman, 1993)

- Define OLS estimate of α as $\hat{\alpha} = (Z^T Z)^{-1} Z^T y = D^{-1} U^T y$.
- Consider the following hierarchical model with $\sigma^2, \tau^2 > 0$:

$$\begin{aligned}(\hat{\alpha}_i | \alpha_i, \sigma^2) &\stackrel{ind}{\sim} \mathcal{N}(\alpha_i, \sigma^2 d_i^{-2}), \\(\alpha_i | \sigma^2, \tau^2, \lambda_i^2) &\stackrel{ind}{\sim} \mathcal{N}(0, \sigma^2 \tau^2 \lambda_i^2).\end{aligned}$$

- Given λ_i and τ , the estimate for β , denoted by $\tilde{\beta}$ is given by:

$$\tilde{\alpha}_i = \frac{\tau^2 \lambda_i^2 d_i^2}{1 + \tau^2 \lambda_i^2 d_i^2} \hat{\alpha}_i, \quad \tilde{\beta} = \sum_{i=1}^n \tilde{\alpha}_i w_i,$$

where $\tilde{\alpha}_i = E(\alpha_i | \tau, \lambda_i^2, X, y)$, w_i is the i th column of the $p \times n$ matrix W and the term $\tau^2 \lambda_i^2 d_i^2 / (1 + \tau^2 \lambda_i^2 d_i^2) \in (0, 1)$ is the shrinkage factor.

Some examples: ridge, PCR and regression with g-prior

- For ridge regression, $\lambda_i^2 = 1$ for all i and we have $\tilde{\alpha}_i = \{\tau^2 d_i^2 / (1 + \tau^2 d_i^2)\} \hat{\alpha}_i$.
- For K component PCR, λ_i^2 is infinite for the first K components and then 0. Thus, $\tilde{\alpha}_i = \hat{\alpha}_i$ for $i = 1, \dots, K$ and $\tilde{\alpha}_i = 0$ for $i = K + 1, \dots, n$.
- For regression with g-prior, $\lambda_i^2 = d_i^{-2}$ and we have $\tilde{\alpha}_i = \{\tau^2 / (1 + \tau^2)\} \hat{\alpha}_i$ for $i = 1, \dots, n$.

Stein's unbiased risk estimate or SURE (Stein, 1981, AoS)

- If “prediction” is the main modeling goal, then the fitted risk is an underestimation of the prediction risk.
- Define the fit $\tilde{y} = X\tilde{\beta} = Z\tilde{\alpha}$, where $\tilde{\alpha}$ is the posterior mean of α .
- Then SURE is given by

$$R = \|y - \tilde{y}\|^2 + 2\sigma^2 \sum_{i=1}^n \frac{\partial \tilde{y}_i}{\partial y_i},$$

where $\sum_{i=1}^n (\partial \tilde{y}_i / \partial y_i)$ is the “degrees of freedom.”

SURE for global shrinkage regressions

- A simple formula need not exist for the degrees of freedom!
- However, since our estimates are posterior means under certain priors, perhaps we can get some simplifications?
- According to Tweedie's formula:

$$\tilde{\alpha} = \hat{\alpha} + \sigma^2 D^{-2} \nabla_{\hat{\alpha}} \log m(\hat{\alpha}).$$

- Noting that $y = Z\hat{\alpha}$ and $\tilde{y} = Z\tilde{\alpha}$ and $\hat{\alpha}_i$ s are independent:

$$R = \sigma^4 \sum_{i=1}^n d_i^{-2} \left\{ \frac{\partial}{\partial \hat{\alpha}_i} \log m(\hat{\alpha}_i) \right\}^2 + 2\sigma^2 \sum_{i=1}^n \left\{ 1 + \sigma^2 d_i^{-2} \frac{\partial^2}{\partial \hat{\alpha}_i^2} \log m(\hat{\alpha}_i) \right\}.$$

SURE for global shrinkage regressions (contd.)

- Thus, calculating the first two derivatives of the log marginal of the **independent** $\hat{\alpha}_i$ s is enough to calculate SURE!
- Integrating out α_i , it is easy to see that

$$(\hat{\alpha}_i \mid \sigma^2, \tau^2, \lambda_i^2) \stackrel{ind}{\sim} \mathcal{N}(0, \sigma^2(d_i^{-2} + \tau^2 \lambda_i^2)).$$

- After elementary calculations, SURE is $R = \sum_{i=1}^n R_i$ where

$$R_i = \frac{\hat{\alpha}_i^2 d_i^2}{(1 + \tau^2 \lambda_i^2 d_i^2)^2} + 2\sigma^2 \frac{\tau^2 \lambda_i^2 d_i^2}{(1 + \tau^2 \lambda_i^2 d_i^2)}.$$

Difficulties with purely global shrinkage

- Recall that in purely global shrinkage λ_i^2 are fixed and there is a single tuning parameter τ .
- If a small τ is chosen $df \approx 0$ but terms with large $\hat{\alpha}_i^2 d_i^2$ make a large contribution to SURE.
- If a large τ is chosen it solves the above problem, but at the expense of a $df \approx 2\sigma^2$ for all terms!
- Maybe **component-specific shrinkage** will help?
- Also note the shrinkage factor $\tau^2 \lambda_i^2 d_i^2 / (1 + \tau^2 \lambda_i^2 d_i^2)$ is monotone in d_i for any given τ and fixed λ_i s.

Global-local shrinkage regression

- Consider the equations

$$\begin{aligned}(\hat{\alpha}_i \mid \alpha_i, \sigma^2) &\stackrel{ind}{\sim} \mathcal{N}(\alpha_i, \sigma^2 d_i^{-2}), \\(\alpha_i \mid \sigma^2, \tau^2, \lambda_i^2) &\stackrel{ind}{\sim} \mathcal{N}(0, \sigma^2 \tau^2 \lambda_i^2), \\ \lambda_i &\stackrel{ind}{\sim} p(\lambda_i).\end{aligned}$$

- The first two equations are the same as before.
- However, now we treat λ_i as random and put a half-Cauchy prior on it, i.e.,

$$p(\lambda_i) \propto \frac{1}{1 + \lambda_i^2}.$$

A bit more on the choice of prior

- The induced prior on α_i on the previous slide is the so called “horseshoe prior.”
- A small τ should help in shrinking the small α_i terms to zero.
- The half-Cauchy prior on λ_i has heavy tails. This should help in “not shrinking” the large α_i terms too much.
- This is what Polson and Scott (2012) did in simulations and noticed good prediction results.
- But can we rigorously show an improved prediction risk estimate?

SURE for global-local shrinkage regression

Theorem 1

Let $m'(\hat{\alpha}_i) = (\partial/\partial\hat{\alpha}_i)m(\hat{\alpha}_i)$ and $m''(\hat{\alpha}_i) = (\partial^2/\partial\hat{\alpha}_i^2)m(\hat{\alpha}_i)$. Then,

- A. SURE for the global-local shrinkage regression model is given by $R = \sum_{i=1}^n R_i$, where

$$R_i = 2\sigma^2 - \sigma^4 d_i^{-2} \left\{ \frac{m'(\hat{\alpha}_i)}{m(\hat{\alpha}_i)} \right\}^2 + 2\sigma^4 d_i^{-2} \frac{m''(\hat{\alpha}_i)}{m(\hat{\alpha}_i)}.$$

- B. Under independent standard half-Cauchy prior on λ_i s, for the second and third terms in Part A we have:

$$\frac{m'(\hat{\alpha}_i)}{m(\hat{\alpha}_i)} = -\frac{\hat{\alpha}_i d_i^2}{\sigma^2} \mathbb{E}(Z_i), \text{ and, } \frac{m''(\hat{\alpha}_i)}{m(\hat{\alpha}_i)} = -\frac{d_i^2}{\sigma^2} \mathbb{E}(Z_i) + \frac{\hat{\alpha}_i^2 d_i^4}{\sigma^4} \mathbb{E}(Z_i^2),$$

where $(Z_i | \hat{\alpha}_i, \sigma, \tau)$ follows a

CCH($p = 1, q = 1/2, r = 1, s = \hat{\alpha}_i^2 d_i^2 / 2\sigma^2, v = 1, \theta = 1/\tau^2 d_i^2$) distribution.

Some remarks on Theorem 1

- The previous theorem establishes that SURE for global-local regression can be expressed by the first two moments of the compound confluent hypergeometric (CCH) distribution.
- These moments can be expressed as doubly infinite series that converge relatively fast and numerical calculations are quick (Gordy, 1998).
- An easy consequence is that now one can do a one-dimensional optimization on τ to minimize SURE.

SURE when $\hat{\alpha}_i^2 d_i^2$ is large and when it is small

Theorem 2

Define $s_i = \hat{\alpha}_i^2 d_i^2 / 2\sigma^2$. When $s_i \gg 1$, both $m''(\hat{\alpha}_i)/m(\hat{\alpha}_i)$ and $[m'(\hat{\alpha}_i)/m(\hat{\alpha}_i)]^2$ are $O(1/\hat{\alpha}_i^2)$ and therefore, the contributions of the second and the third terms to R_i is $O(1/\hat{\alpha}_i^2 d_i^2)$. Consequently, the component-wise SURE $R_i \approx 2\sigma^2$.

Theorem 3

Define $s_i = \hat{\alpha}_i^2 d_i^2 / 2\sigma^2$. Then the following statements are true.

- A. The component-wise SURE R_i is an increasing function of s_i in the interval $[0, 1]$ for any fixed τ .
- B. When $s_i = 0$, the component-wise SURE R_i is a monotone increasing function of τ , and is bounded in the interval $(0, 2\sigma^2/3]$ when $\tau^2 d_i^2 \in (0, 1]$.

Some remarks on Theorems 2 and 3

- Recall that SURE for pure global regression $R = \sum_{i=1}^n R_i$ where

$$R_i = \frac{\hat{\alpha}_i^2 d_i^2}{(1 + \tau^2 \lambda_i^2 d_i^2)^2} + 2\sigma^2 \frac{\tau^2 \lambda_i^2 d_i^2}{(1 + \tau^2 \lambda_i^2 d_i^2)}.$$

- For global-local regression, Theorem 2 establishes that the terms with $s_i = \hat{\alpha}_i^2 d_i^2 / 2\sigma^2 \gg 1$ will contribute $2\sigma^2$ to SURE.
- For global-local regression, Theorem 3 establishes that terms with $s_i = 0$ contribute less than $2\sigma^2/3$ to SURE, provided τ is chosen sufficiently small, i.e., $\tau^2 \leq d_i^{-2}$.
- Simultaneously controlling SURE in these two situations (i.e., $s_i \gg 1$ and $s_i = 0$) is not possible with a single τ .*

Numerical examples

Table : The true orthogonalized regression coefficients α_{0i} , their OLS estimates $\hat{\alpha}_i$, and singular values d_i of X , for $n = 100$ and $p = 500$.

i	α_{0i}	$\hat{\alpha}_i$	d_i	$\hat{\alpha}_i d_i$
1	0.10	0.10	635.10	62.13
2	-0.44	-0.32	3.16	-1.00
...	...	...	...	...
5	-0.13	0.30	3.05	0.91
6	10.07	10.22	3.02	30.88
...	...	...	...	...
29	0.46	0.60	2.53	1.53
30	10.47	11.07	2.51	27.76
...	...	...	...	...
56	0.35	0.57	2.07	1.18
57	10.23	10.66	2.07	22.05
...	...	...	...	...
66	-0.00	-0.35	1.90	-0.66
67	11.14	11.52	1.88	21.70
...	...	...	...	...
95	-0.82	-0.56	1.42	-0.79
96	9.60	10.21	1.40	14.26
...	...	...	...	...
100	0.61	0.91	1.27	1.15

Numerical examples (contd.)

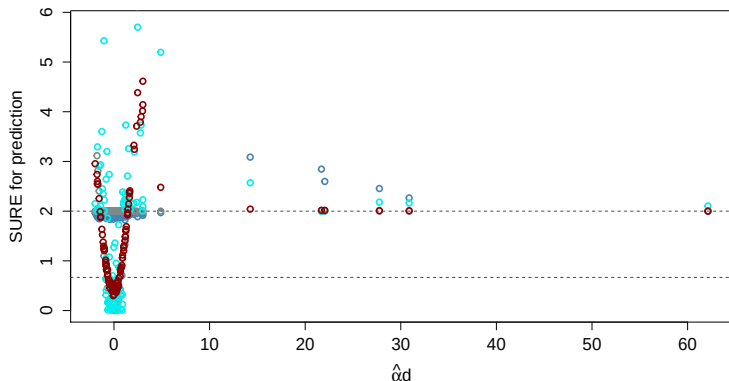


Figure : SURE for ridge (blue), PCR (gray), lasso (cyan) and horseshoe regression (red), versus $\hat{\alpha}d$, where $\hat{\alpha}$ is the OLS estimate of the orthogonalized regression coefficient, and d is the singular value, for $n = 100$ and $p = 500$. Dashed horizontal lines are at $2\sigma^2 = 2$ and $2\sigma^2/3 = 0.67$.

Numerical examples (contd.)

Table : SURE and average out of sample prediction SSE (standard deviation of SSE) on one training set and 200 testing sets for the competing methods for $n = 100$. The lowest SURE in each row is in blue and the lowest average prediction SSE is in red.

p	RR		LASSO		A.LASSO	PCR		HS	
	SURE	SSE	SURE	SSE	SSE	SURE	SSE	SURE	SSE
100	159.02	168.24 (23.87)	125.37	128.98 (18.80)	127.22 (18.10)	162.23	179.81 (25.51)	120.59	126.33 (18.77)
200	187.38	174.92 (21.13)	140.99	132.46 (18.38)	151.89 (20.47)	213.90	191.33 (22.62)	139.32	126.99 (17.29)
300	192.78	191.91 (22.95)	147.83	145.04 (19.89)	153.64 (21.19)	260.65	253.00 (26.58)	151.24	136.67 (18.73)
400	195.02	182.55 (22.70)	148.56	165.63 (21.55)	178.98 (20.12)	346.19	292.02 (28.98)	147.69	143.91 (18.41)
500	196.11	188.78 (22.33)	159.95	159.56 (19.94)	186.23 (23.50)	386.50	366.88 (39.38)	144.97	160.11 (20.29)

Concluding remarks

Global-local priors: originally designed for sparse normal means model.

Seem to **work well for default Bayes analysis** e.g. the Efron problems.

- Some theoretical insight is provided by Bhadra et al. (2016).
- Much work still remains to be done for a rigorous justification.

Seem to **provide improved prediction risk** in regression.

- Optimality results?

References

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